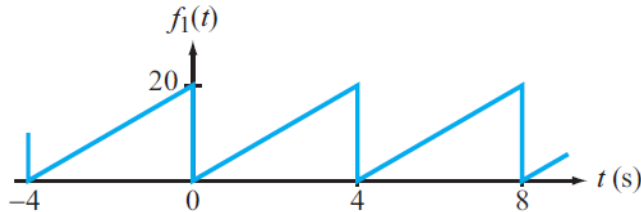


1. (60 points) Calculate the three Fourier series representations for the following waveform, as requested below.



- (20 points) Write down the sine/cosine representation of the Fourier series for the above waveform
- (20 points) Write down the amplitude/phase representation for the above waveform
- (20 points) Write down the exponential representation for the above waveform
- (20 points) Use the MATLAB code given below to generate the sine/cosine representation of the Fourier series for the above waveform (also posted as FourierSeries.m on canvas)

```
%
% ECE 3337 Signals and Systems Analysis
% University of Houston
% Calculate the Fourier series coefficients for a signal x(t)
% x(t) can be typed in as a formula
% This example uses the MATLAB symbolic Toolbox
clear; % always start by clearing the workspace
% Define the symbols
syms t a0 a(k) b(k) x(t) y(t) w0 T0 A;
syms n k N integer;
assume(T0>0);
w0 = 2*pi/T0; % angular frequency
% Textbook square-wave example
x(t) = piecewise(0<=t<T0/4,A,T0/4<t<3*T0/4,-A,3*T0/4<t<=T0,A);
% calculate the Fourier coefficients
a0 = simplify((1/T0)*int(x(t),t,0,T0));a0;
a(k) = simplify((2/T0)*int(x(t)*cos(k*w0*t),t,0,T0));
b(k) = simplify((2/T0)*int(x(t)*sin(k*w0*t),t,0,T0));
% Write out the first N terms of the Fourier series
N = 10; % number of terms of the Fourier series to be written out
y(t) = a0 + symsum(a(k)*cos(k*w0*t),k,1,N) + symsum(b(k)*sin(k*w0*t),k,1,N)
```

- (30 points) The signal $x(t)$ with $A = 10\text{V}$ and $T = 1\text{ms}$ is applied as an input to the following circuit.
 - (15 points) Derive the Fourier series for the output signal $v_{out}(t)$ taking advantage of table 5-4 of your textbook
 - (15 points) Calculate the first five terms of $v_{out}(t)$ when $R_1 = 1\Omega$, $C = 1\text{F}$, $L = 1\text{H}$

